AI 101 for Mechanical Engineers · Module 11

Batches, epochs and realistic expectations

Lesson 11.3

What you will learn

Distinguish batch, update and epoch, calculate updates per epoch, explain why more training does not guarantee better unseen performance, and link validation checkpoints with a final test.

Lesson overview

Module 11 · Lesson 3

Batches, epochs and realistic expectations

AI 101 for Mechanical Engineers

1 Batch, update and epoch

Training set: 1,000 examples in 20 batches of 50



Batch a group of examples (orange) used for one update

Update one change of the parameters from that batch's gradients

Epoch one complete pass through all the training data

Epochs count passes; updates count parameter changes. All are counts, with no SI unit.

2 Updates per epoch

$$U = N_{total} / N_{batch}$$

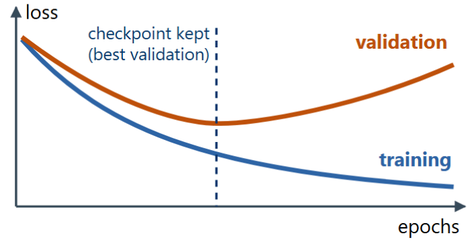
N_{total} = 1,000; N_{batch} = 50

U = 1,000 ÷ 50 **= 20**

Exact here. If the data do not divide evenly, state how the last partial batch is handled.

Larger batches: less noisy gradient estimate, fewer updates per epoch. Smaller batches: noisier, more updates. Evaluate; neither is always better.

3 More epochs, not always better



Illustrative curves, not measured results; real curves vary.

- Training loss can fall while validation loss rises
- An epoch does not guarantee lower training loss
- Select the checkpoint with validation data; use the locked test only after selection

Key takeaway

More training is not automatically better.

Count updates correctly, choose the checkpoint with validation data, and keep the final test separate.

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A batch is a group; an update changes parameters

Batch

- Group of training examples for an update [4]
- Batch size is a count, no SI unit

Update

- Changes parameters using gradient information
- Follows predictions, loss and gradients

An epoch is one complete pass

- One epoch can contain several parameter updates
- Epochs count full passes; updates count parameter adjustments
- An epoch does not guarantee lower training loss [4, Ch. 8]
- Step size, example order and optimizer affect the result [4, Ch. 8]

1,000 examples in batches of 50: 20 updates

$$U = N_{total} / N_{batch}$$

$$U = 1{,}000 \div 50 = 20$$

- All quantities are counts with no physical units
- Exact division: 20 batches and 20 updates per epoch
- Otherwise, state how the final partial batch is treated

Training and validation loss can diverge

Training loss

- Keeps decreasing as training continues

Validation loss

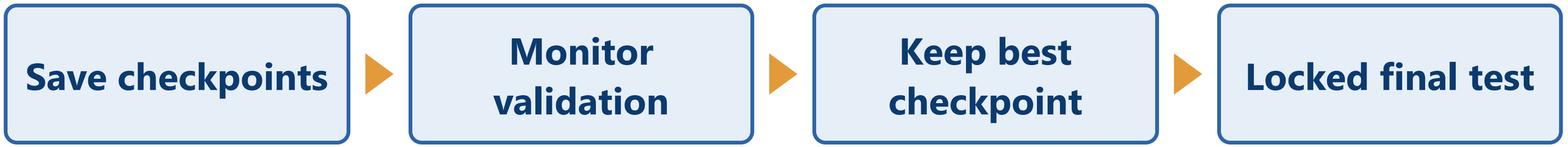
- Decreases at first, then increases

Illustrative curves: not measured machinery data, no stopping epoch, shape may differ each run.

More epochs do not guarantee better generalization

- Deep networks typically rely on large datasets and computation [3]
- Their performance must be checked on held-out data
- Extra epochs can fit training-specific variation
- Lower training loss does not guarantee better test performance

Select a checkpoint, then test separately



- Early stopping keeps the copy with best validation error [56]
- Do not assume the last epoch is best
- Keep final-test results out of checkpoint selection

Keep expectations introductory

- Goal: understand neural computations and training vocabulary
- No GPU is required for this course
- Full architecture design and large-scale training are later study
- Synthetic examples do not replace physical equipment validation

Batch size is a setting to evaluate

Larger batch

- Usually a less noisy gradient estimate [4, Sec. 8.1.3]
- Fewer updates per epoch

Smaller batch

- Noisier updates [4, Sec. 8.1.3]
- More updates per epoch

Effects on speed and generalization depend on model, data and optimizer.

CHECK YOUR UNDERSTANDING

How many updates occur in one standard epoch with 1,000 examples and batches of 50?

Answer before moving on.

CHECK YOUR UNDERSTANDING

How many updates occur in one standard epoch with 1,000 examples and batches of 50?

> 20 updates, because 1,000 ÷ 50 = 20. The example assumes an exact division into full batches.

CHECK YOUR UNDERSTANDING

Would 100 additional epochs guarantee better test performance?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Would 100 additional epochs guarantee better test performance?

> No. Extra training can fit training-specific variation. Validation guides checkpoint selection; a locked final test then estimates performance.

Key takeaways

- A batch feeds an update; an epoch is one full pass
- Updates per epoch depend on dataset size, batch size, partial batches
- More epochs do not guarantee better generalization
- Select checkpoints by validation, then use the locked final test

References

[3] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” Nature, vol. 521, no. 7553, pp. 436–444, 2015. doi: 10.1038/nature14539

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.

[56] L. Prechelt, “Early stopping—But when?,” in Neural Networks: Tricks of the Trade (Lecture Notes in Computer Science, vol. 1524), G. B. Orr and K.-R. Müller, Eds. Berlin, Germany: Springer, 1998, pp. 55–69. doi: 10.1007/3-540-49430-8_3

Numbers match the course reference list. The written lesson is the authoritative text.